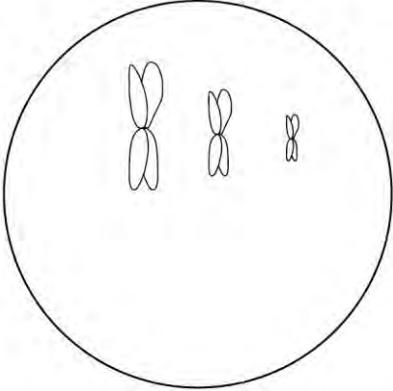


Question	Marking Guidance	Mark	Comments
07.1	Correct answer of $5.859375 \times 10^4 = 2$ marks;; $2^6 / 64$ (correct number of cells from 6 mitotic cycles) = 1 mark OR 2.34×10^5 or 234375 (correctly dividing 1.5×10^7 by 64) = 1 mark OR $2^8 / 256$ (correct number of sperm cells produced from one A₁ cell) = 1 mark OR Correct answer not in standard form, eg 58594 = 1 mark;	2 (2 x AO2)	Accept, correct rounding of 5.859375×10^4, eg $5.9 \times 10^4 / 6 \times 10^4$ = 2 marks Accept, correct rounding of $2.34 \times 10^5 / 234375 = 1$ mark Accept, correct rounding of 58593.75 = 1 mark

Question	Marking Guidance	Mark	Comments
07.2	1. Correct labels with cell phases on the x-axis, and DNA mass per cell with arbitrary units/au on y-axis with linear scale and bars not touching; 2. (Early interphase plotted at) 8 and (late interphase at) 16; 3. (End of 1st meiotic division plotted at) 8 and (end of 2nd meiotic division at) 4;	3 (3 x AO3)	1. Reject if Y axis does not cover at least half of the grid 1. Accept appropriate labels OR key and DNA mass in a/one cell If 2 and 3 not achieved, accept for 1 mark, bars showing halving of DNA content during meiosis

Question	Marking Guidance	Mark	Comments
07.3	Accept for 1 mark, 3 chromosomes, each of different size and each with 2 chromatids, drawn in any order; For example, 	1 (1 x AO1)	

Question	Marking Guidance	Mark	Comments
07.4	1. Non-disjunction; 2. Homologous chromosomes/pairs not separated OR Chromatids not separated; 3. (Because) both chromosome(s)/chromatid(s) pulled/moved to (one) pole/end/side (of cell) OR (Because) both chromosome(s)/chromatid(s) not attached to spindle (fibres);	3 max (3 x AO1)	1. Accept trisomy OR aneuploidy 1. Reject polyploidy/multiple sets 2. Accept segregated for 'separated' 2. Accept 'bivalent' for 'homologous chromosomes' 3. Accept labelled diagram showing chromosome /chromatid movement to one pole/cell
